1. Numbers

1.1. Types of Numbers

- Natural Numbers
  - They start from 1
  - Used for counting purposes
  - All positive integers excluding zero
- Integer
  - All whole numbers: positive, negative and zero.
  - For eg: $0, \pm1, \pm2, \pm3$
- Prime Numbers
  - A number which has exactly **two distinct factors**: itself and 1
  - 1 is **not** a prime number
- Rational Numbers
  - Any number that can be written as a fraction p/q, where p and q are integers and q ≠ 0.
- Irrational Numbers
  - Numbers that **cannot** be written as simple fractions.
  - Their decimals are **non-terminating** and **non-recurring**, e.g. π
- Square Numbers
  - The result of multiplying a number by itself twice
  - For eg, 1,4,9,16,25,36,49,64,81,100
- Cube Numbers
  - Made from multiplying a number by itself three times. For eg, 1,8,27,64,125
- Reciprocals
  - The reciprocal of a number is **1 divided by that number**
  - For eg, Reciprocal of 4 is ¼
  - Zero has **no reciprocal**
- Common Multiples
  - A number that is a multiple of two or more given numbers
  - For eg, Common Multiples of 6 and 8 are 24 and 48, as they appear common in both tables of 6 and 8
- Common factors
  - A number that divides two or more numbers exactly
  - For eg,

    Factors of 12 : 1, 2 , 3 , 4 , 6 and 12

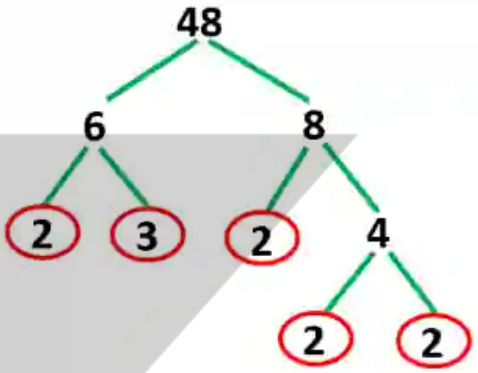
    Factors of 18 : 1, 2 , 3 , 6 , 9 and 18

    Common factors of 12 and 18: 1, 2, 3 and 6

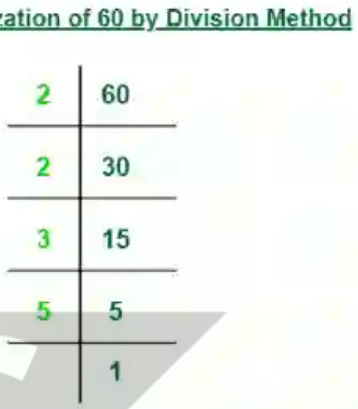
1.2. Prime Factors, LCM, and HCF

Prime Factorisation

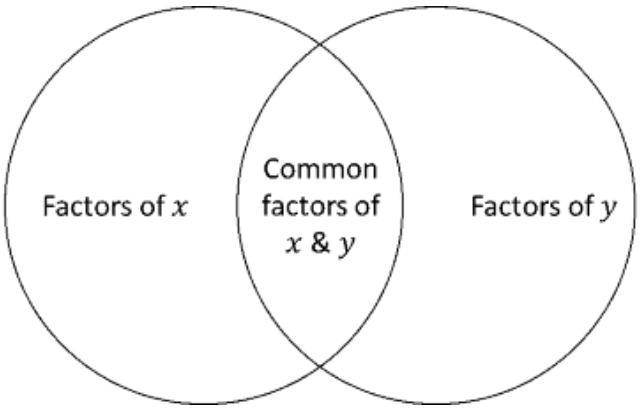
- Writing a number as a product of its prime factors
- For eg, 60 = 2^2 x 3 x 5
- Methods:
  - **Factor Tree Method** - break the number down step-by-step into primes



  - **Division Method** - keep dividing by the smallest prime until you reach 1



Highest Common Factor and Lowest Common Multiple



- **HCF:** Product of common prime factors

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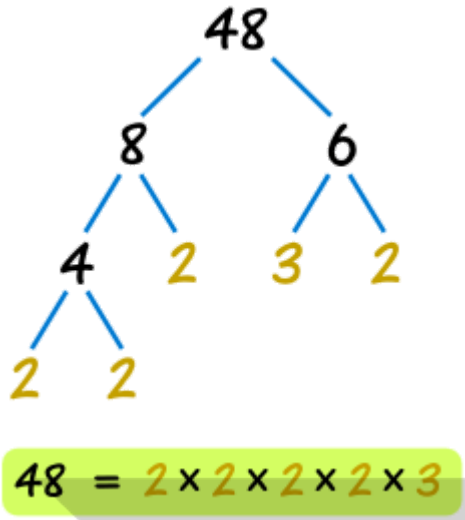
- **LCM:** Product of all prime factors in the Venn Diagram



1.3. Power and Roots

Squares and Square Roots

- **Square** of a number means multiplying it by itself.
  - $n^2 = n \times n$
- **Square Root** is the number that, when squared, gives the original number
  - $3^2 = 9$, so the square root of 9 is 3
  - $(-3)^2 = 9$, so both +3 and -3 are square roots of 9.
  - $\sqrt{n^2} = n$
- **Examples:**
  - $4^2 = 16$
  - $\sqrt{25} = 5$
  - $(-7^2) = 49$
  - $\sqrt{169} = 13$

Cube and Cube Roots

- **Cube of a number** means multiplying it by itself three times:
  - $n^3 = n \times n \times n$
- **Cube Root** is the number that, when cubed, gives the original number
  - $\sqrt[3]{n^3} = n$
- **Examples:**
  - $2^3 = 8$
  - $\sqrt[3]{27} = 3$
  - $(-3)^3 = -27$
  - $\sqrt[3]{-64} = -4$

Other Powers and Roots

- Numbers can have **fractional**, **negative**, or **higher powers**
- A **power** tells you how many times to multiply a number by itself.
  - **Example:** $2^4 = 2 \times 2 \times 2 \times 2 = 16$
- A **root** is the inverse of a power; it 'undoes' the power
  - **Example:** $\sqrt[4]{16} = 2$, because $2^4 = 16$
- The nth root of a number x is the value which, when raised to the power of n, gives x.
  - $\sqrt[n]{x} = y$, $y^n = x$
- **Fractional Indices** represent roots and powers together
  - $x^{\frac{1}{2}} = \sqrt{x}$
  - $x^{\frac{1}{3}} = \sqrt[3]{x}$
  - $x^{\frac{m}{n}} = \sqrt[n]{x^m}$

1.4. Estimation and Accuracy

Accuracy

- Upper Bound - The largest possible value a measurement could be, given its degree of accuracy.
- Lower Bound - The smallest possible value a measurement could be, given its degree of accuracy.
- Worked Example

  Q. Find the upper and lower bound of 450 to the nearest 10

  Half the degree of accuracy = $\frac{10}{2}$ = 5

  Add to original value for upper bound = 450 + 5 = 455

  Subtract from original value for lower bound = 450 - 5 = 445

Estimation

Significant Figures

- Significant figures is how precise a number is

Rules for Significant Figures

- All non-zero digits count
  - 348 → 3 s.f
- Zeros between digits count
  - 406 → 3 s.f
- Leading zeroes do not count
  - 0.54 → 2 s.f
- Trailing zeros count only if there is a decimal
  - 1500 (no decimal) → 2 s.f
  - 12.30 → 4 s.f

Examples

- Round to 1 s.f
  - 3.76 → 4
  - 248 → 200
- Round to 2 s.f
  - 3.76 → 3.8
  - 248 → 250

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1.5. Sets

Definition of Sets

- A set is a collection of elements. e.g.
  - $A = x : x \text{ is a natural number}$ [A is the set of all natural numbers]
  - $B = (x, y) : y = mx + c$
  - $C = x : a \leq x \leq b$
  - $D = a, b, c, \ldots$

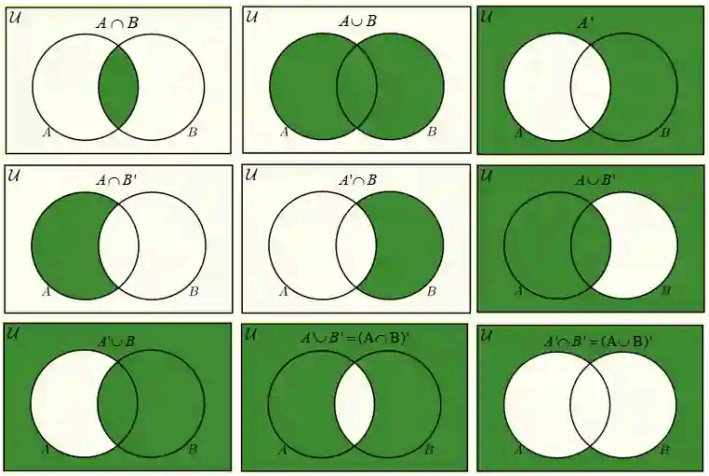
Set Notation

- $\in$ = ...is an element of...
- $\notin$ = ...is not an element of...
- A' = complement of set A
  - The complement of Set A is the set of all the elements that are inside the universal set but not in Set A
- $\emptyset$ = empty set (e.g. A = {})
- ξ = Universal set
  - A set that contains all elements of all other sets
- $\cup$ = Union
  - All elements in A and B
- $\cap$ = Intersection
  - Common elements in A and B
- $\subseteq$ = Subset
  - All elements of A are in B, and A can be = B
- $\subset$ = Proper Subset
  - All elements of A are in B, and A ≠ B
- $\not\subseteq$ = Not a Subset
  - A has some element outside of B.
- n(A) - the number of elements in a set A

Equal Sets

- Two sets that contain the same elements, regardless of the order in which the elements appear

Set Representations



1.6. Fractions, Decimals, and Percentages

Fractions

- Used to express a part of a whole. eg, $\frac{2}{5}$
- **Numerator**: The numerator is how many parts you have
- **Denominator**: The denominator is how many parts the whole is divided into
- **Proper Fraction**: When the numerator is less than the denominator. For example, $\frac{3}{8}$
- **Improper Fraction**: When the numerator is greater than the denominator. For example, $\frac{5}{3}$
- **Mixed Fractions**
  - This is only possible for improper fractions
  - A mixed fraction is a whole number with a proper fraction
  - Conversion of an Improper Fraction to a Mixed Fraction:

    The remainder becomes the numerator of the proper fraction, with the denominator remaining the same.

    $\frac{9}{4} = 2\frac{1}{4}$

Decimals

- Consists of a whole number part and a fractional part separated by a decimal point
- **Recurring Decimals:** A digit repeating infinitely after the decimal point (e.g $7.\overline{3}$,which indicates 7.33333....)
- **Recurring Decimals to Fraction :**
  - Let x = recurring decimal
  - Let n = number of recurring digits
  - Multiply the recurring decimal by 10^n
  - Subtract the original recurring decimal from the product above.
  - Solve for x, expressing your answer in the simplest form

Percentages

- Per cent means 'out of a hundred', referring to values out of 100

Conversions of Fractions, Decimals, and Percentages

- To convert a Percentage to a Decimal
  - Divide by 100

    $45\% = \frac{45}{100} = 0.45$
- To convert a Decimal to a Percentage
  - Multiply by 100

    $0.7 \times 100 = 70\%$
- To convert a Percentage to a Fraction
  - Write the fraction with a denominator of 100 and simplify

    $25\% = \frac{25}{100} = \frac{1}{4}$

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- To convert a Fraction to a Percentage
  - Divide the numerator by the denominator and multiply by 100

$\frac{13}{29} = 0.4482, \quad 0.4482 \times 100 = 44.82\%$

1.7. Ordering

- $=$ shows when 2 values are equal
- $\neq$ when 2 values are not equal
- $>$ when the value on the left-hand side is strictly greater than the right-hand side
- $<$ when the value on the left-hand side is strictly less than the right-hand side
- $\geqslant$ when the value on the left-hand side is greater than or equal to the right-hand side
- $\leqslant$ when the value on the left-hand side is less than or equal to the right-hand side

1.8. Indices Rules

- When two powers with the same base are multiplied, the values are added

$$2^3 \times 2^4 = 2^{3+4} = 2^7$$

- When two powers with the same base are divided, the values are subtracted:

$$2^5 \div 2^2 = 2^{5-2} = 2^3$$

- Any number raised to the power of 0 equals 1:

$$5^0 = 1$$

- When a power is raised to another power, the values are multiplied:

$$(3^2)^3 = 3^{2\times3} = 3^6$$

- A negative power means the reciprocal of the base raised to the positive power:

$$4^{-2} = \frac{1}{4^2} = \frac{1}{16}$$

- A fractional power indicates a root; the denominator is the root, and the numerator is the power:

$$8^{\frac{2}{3}} = \sqrt[3]{8^2} = \sqrt[3]{64} = 4$$

- When two numbers that share the same exponent, they're able to share their operation and be raised to the same power.

$2^2 \times 3^2 = (2 \times 3)^2$

$2^2 \div 3^2 = \left(\frac{2}{3}\right)^2$

1.9. Standard Form

- Used to express large or small numbers in factors of 10

To Express Numbers Greater than 0

- Let us take the example of the number 450,000
- Count the number of digits after the first digit, in this case, 5
- Take that as the power of 10
- Place a decimal point after the first number and remove any non-significant 0's
- $450,000 = 4.5 \times 10^5$

To Express Numbers Less than 0

- Let us take the example of the number 0.00031
- Count the number of digits before the first significant digit, in this case, 4
- Take that as the negative power of 10
- Place a decimal point after the first number and remove any non-significant 0's
- $0.00031 = 3.1 \times 10^{-4}$

Ratio Simplification

- Find the HCF of the numbers in the ratio
- Divide both by the HCF
- $21{:}35$
- HCF = 7
- $3{:}5$

Surds

Rules of Surds

- $\sqrt{ab} = \sqrt{a}\sqrt{b}$
  - $\sqrt{50} = \sqrt{25 \times 2} = 5\sqrt{2}$
- $\sqrt{a} \times \sqrt{b} = \sqrt{ab}$
  - $\sqrt{3} \times \sqrt{12} = \sqrt{36} = 6$
- $\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}$
  - $\frac{\sqrt{27}}{\sqrt{3}} = \sqrt{\frac{27}{3}} = \sqrt{9} = 3$
- You can only add or subtract **like** surds
  - $3\sqrt{5} + 2\sqrt{5} = 5\sqrt{5}$
  - $\sqrt{2} + \sqrt{3} \rightarrow$ do not combine

1.10. Ratio & Proportion

Ratio

- Used to describe a fraction
  - e.g. 3:1

Foreign Exchange

- Money changed from one currency to another using proportion
- E.g., Convert \$22.50 to Dinars

\$1 : 0.30KD
\$22.50 : 6.75KD

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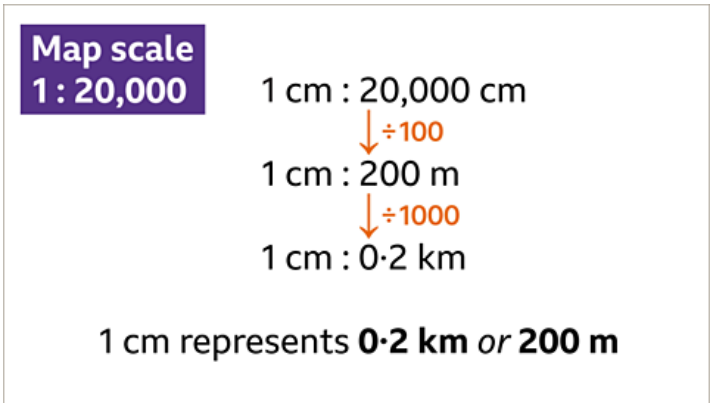
Map Scales

- Using proportion to work out map scales



- 1km = 1000m
- 1m = 100cm
- 1cm = 10mm

Direct Variation

- y is proportional to x

$y \propto x$

$y = kx$

Inverse Variation

- y is inversely proportional to x

$y \propto \frac{1}{x}$

$y = \frac{k}{x}$

1.11. Percentages

- Percentages are a method of showing the ratio between the number and the max out of 100
- To express one quantity as a percentage of another, divide the first quantity by the second, then multiply by 100.

$$\frac{50}{200} \times 100 = 0.25 \times 100 = 25\%$$

- To find a percentage of a quantity, multiply the quantity by the percentage (as a decimal).

$$15\% \times 200 = \frac{15}{100} \times 200 = 0.15 \times 200 = 30$$

- To calculate the percentage increase or decrease in a value, you can simply do:

$$\text{Percentage Change} = \frac{\text{New Value} - \text{Original Value}}{\text{Original Value}} \times 100$$

- If the result is positive, it's a **percentage increase**.
- If the result is negative, it's a **percentage decrease.**
- **How to Increase/Decrease a number by a Percentage**

$$\left(\frac{100 \pm \text{percentage change}}{100}\right) \times \text{Original value}$$

- This formula is also used for Exponential growth (increase) and decay (decrease)

Simple Interest

- This is when a value is increased based on the initial start amount (Principal Amount)
- You can use this formula to calculate the interest amount (You need to add the principal amount again to find the final amount)

$$\text{Simple Interest} = \frac{P \times R \times T}{100}$$

$$\text{P} = \text{Principal Amount}$$

$$\text{R} = \text{Rate}$$

$$\text{T} = \text{Time}$$

Compound Interest

- Compound interest is when the interest is based on the latest value
- You can use this formula to calculate the interest amount

$$\text{Compound Interest} = \text{Principal} \times \left(1 + \frac{\text{Rate}}{100}\right)^{\text{Time}}$$

Reverse Percentage

1.12. Time and Currency

Time Conversions

- 1 Minute = 60 seconds
- 1 Hour = 60 minutes
- 1 Year = 365 days (or 366 in a leap year).
- 1 Day = 24 Hours
- 1 Week = 7 Days
- 1 Month = 30/31 Days
- If you are asked to convert 3 hours into minutes:

$3 \text{ hours} \times 60 \text{ minutes} = 180 \text{ minutes}.$

- The **12-hour clock** uses "a.m." for morning and "p.m." for afternoon/evening.
- The **24-hour clock** uses a continuous format from 00:00 (midnight) to 23:59.

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- Steps to convert 12-hour to 24-hour format:
  - For **a.m.**, keep the time as is but add leading zeros if needed (e.g., $3:15 \text{ a.m.} = 03:15$)
  - For **p.m.**, add 12 to the hour (e.g., $3:15 \text{ p.m.} = 15:15$)

Time Zones

- If it is 10 a.m. in London (GMT) and New York is 5 hours behind, the time in New York is:
  $10:00 - 5 \text{ hours} = 5:00 \text{ a.m.}$

Currency Conversion

$$\text{Amount} = \frac{\text{New Rate}}{\text{Old Rate}}$$

$$\text{Amount in Currency B} = \text{Amount in Currency A} \times \text{Exchange Rate}$$

1.13. Speed, Distance & Time

$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$

$\text{Average Speed} = \frac{\text{Total Distance Travelled}}{\text{Total Time Taken}}$

- **Units of Speed:** km/hr or m/s .
- **Units of Distance:** km or m
- **Units of Time:** hr or sec

$$km/hr \times \frac{5}{18} = m/sec$$

$$m/sec \times \frac{18}{5} = km/hr$$

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